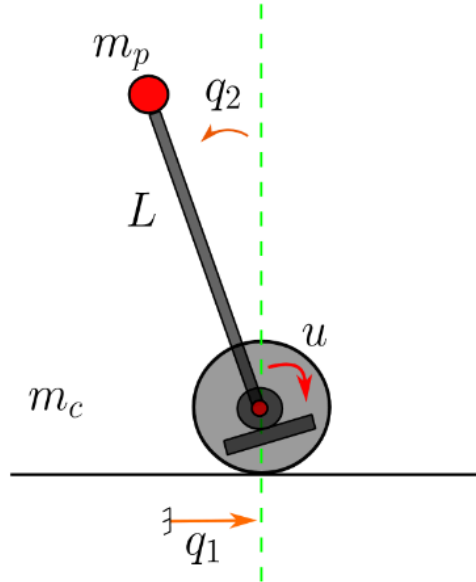


Pandulum-Cart System

A mass m_p (that will be referred to as m) is attached to a car of mass m_c (M) through a massless junction with a length L (l). The car displacement is induced by a force u .



The goal of the system is to stabilize the mass m_p vertically, keeping the swing angle $q_2 = 0$ by controlling the displacement q_1 of the car.

This exercise will cover the following points:

1. Modeling the system using Lagrangian formalism
2. Linearizing the model around a functioning point of $q_2=0$, to obtain a linear state space model
3. Build a Simulink model of the system (linear and non linear model).
4. Design a state feedback controller to stabilize the system
5. Design a optimal controller
6. Design a mpc controller

1. Model of the system

The system has 2 degrees of freedom q_1 and q_2 .

Considering the (x,y) frame the position of the masses are given as

$$\begin{cases} \overrightarrow{OM} = q_1 \vec{x} \\ \overrightarrow{Om} = (q_1 - l \sin(q_2)) \vec{x} + l \cos(q_2) \vec{y} \end{cases}$$

The velocities are:

$$\begin{cases} \frac{d \overrightarrow{OM}}{dt} = \dot{q}_1 \overrightarrow{x} \\ \frac{d \overrightarrow{Om}}{dt} = (\dot{q}_1 + l \dot{q}_2 \cos(q_2)) \overrightarrow{x} - l \dot{q}_2 \sin(q_2) \overrightarrow{y} \end{cases}$$

The cinetic energy of the system is given as

$$T = \frac{1}{2} M \times v_M^2 + \frac{1}{2} m \times v_m^2 = \frac{1}{2} M \dot{q}_1^2 + \frac{1}{2} m \left((\dot{q}_1 + l \dot{q}_2 \cos(q_2)) \overrightarrow{x} - l \dot{q}_2 \sin(q_2) \overrightarrow{y} \right)^2$$

$$T = \frac{1}{2} (M + m) \dot{q}_1^2 + \frac{1}{2} m (l^2 \dot{q}_2^2 - 2l \dot{q}_1 \dot{q}_2 \cos(q_2))$$

The potential energy of the system is produce only by the mass m as the mass M is already grounded. So we have

$$V = mgL \cos(q_2)$$

From the about energies equation, we obtain the Lagrangian

$$L = T - V = \frac{1}{2} (M + m) \dot{q}_1^2 + \frac{1}{2} m (l^2 \dot{q}_2^2 - 2l \dot{q}_1 \dot{q}_2 \cos(q_2)) - mgL \cos(q_2)$$

As the system is not only conservative. It also have dissipative force (friction), generalized Lagrange equations apply as follows:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i$$

Rayleigh dissipative function will be used to introduce the friction force proportionnal to velocity.

$$R(\dot{q}) = \frac{1}{2} \sum_i^N k_i \dot{q}_i^2$$

the dissipative force is given as $F_i = - \frac{\partial R}{\partial \dot{q}_i}$

Thus $Q_i = F_i + < \text{other external forces applied to particle } i >$

- Lagrange equation w.r.t. q_1

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_1} \right) - \frac{\partial L}{\partial q_1} = -k_1 \dot{q}_1$$

$$\frac{\partial L}{\partial \dot{q}_1} = (M + m) \dot{q}_1 - m l \dot{q}_2 \cos(q_2) ;$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_1} \right) = (M + m) \ddot{q}_1 - m l \ddot{q}_2 \cos(q_2) + m l \dot{q}_2^2 \sin(q_2)$$

$$\frac{\partial L}{\partial \dot{q}_1} = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_1} \right) - \frac{\partial L}{\partial q_1} = -k \dot{q}_1 \Rightarrow (M + m) \ddot{q}_1 - m l \ddot{q}_2 \cos(q_2) + m l \dot{q}_2^2 \sin(q_2) + k_1 \dot{q}_1 = 0$$

• Lagrange equation w.r.t. q_2

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_2} \right) - \frac{\partial L}{\partial q_2} = Q_2 \quad . \text{ The virtual work } \delta W \text{ is produced when there is a rotation } \delta q_2 \text{ of the arm. So we}$$

have $\delta W = u \delta q_2$, then $Q_2 = u$

$$\Rightarrow -m l \ddot{q}_1 \cos(q_2) + m l^2 \ddot{q}_2 - m g l \sin(q_2) = -k_2 \dot{q}_2 + u$$

The model of the system is given as

$$\begin{cases} (M + m) \ddot{q}_1 - m l \ddot{q}_2 \cos(q_2) + m l \dot{q}_2^2 \sin(q_2) + k_1 \dot{q}_1 = 0 \\ -m l \ddot{q}_1 \cos(q_2) + m l^2 \ddot{q}_2 - m g l \sin(q_2) + k_2 \dot{q}_2 = u \end{cases}$$

2. Model linearization

$$q_2 \approx 0 \Rightarrow \begin{cases} \cos(q_2) \approx 1 \\ \sin(q_2) \approx q_2 \end{cases}$$

$$\begin{cases} (M + m) \ddot{q}_1 - m l \ddot{q}_2 + k_1 \dot{q}_1 = 0 \\ -m l \ddot{q}_1 + m l^2 \ddot{q}_2 - m g l q_2 + k_2 \dot{q}_2 = u \end{cases}$$

$$\begin{cases} \ddot{q}_1 = \frac{m l}{M + m} \ddot{q}_2 - \frac{k_1}{M + m} \dot{q}_1 \\ \ddot{q}_2 = \frac{1}{l} \ddot{q}_1 - \frac{k_2}{m l^2} \dot{q}_2 + \frac{g}{l} q_2 + \frac{1}{m l^2} u \end{cases} \Rightarrow \begin{cases} \ddot{q}_1 = \frac{m}{M + m} \ddot{q}_1 - \frac{k_2}{(M + m)l} \dot{q}_2 + \frac{m g}{(M + m)} q_2 + \frac{m}{(M + m)l} u - \frac{k_1}{M + m} \dot{q}_1 \\ \ddot{q}_2 = \frac{m}{M + m} \ddot{q}_2 - \frac{k_1}{(M + m)l} \dot{q}_1 - \frac{k_2}{m l^2} \dot{q}_2 + \frac{g}{l} q_2 + \frac{1}{m l^2} u \end{cases}$$

$$\Rightarrow \begin{cases} M \ddot{q}_1 = -k_1 \dot{q}_1 - \frac{k_2}{l} \dot{q}_2 + m g q_2 + \frac{m}{l} u \\ M \ddot{q}_2 = -\frac{k_1}{l} \dot{q}_1 - \frac{(M + m) k_2}{m l^2} \dot{q}_2 + \frac{(M + m) g}{l} q_2 + \frac{(M + m)}{m l^2} u \end{cases} \Rightarrow \begin{cases} \ddot{q}_1 = -\frac{k_1}{M} \dot{q}_1 - \frac{k_2}{M l} \dot{q}_2 + \frac{m g}{M} q_2 + \frac{m}{M l} u \\ \ddot{q}_2 = -\frac{k_1}{M l} \dot{q}_1 - \frac{(M + m) k_2}{M m l^2} \dot{q}_2 + \frac{(M + m) g}{M l} q_2 \end{cases}$$

$$\begin{cases} \dot{q}_1 &= \dot{q}_1 \\ \dot{q}_2 &= \dot{q}_2 \\ \ddot{q}_1 &= -\frac{k_1}{M} \dot{q}_1 - \frac{k_2}{M l} \dot{q}_2 + \frac{m g}{M l} q_2 + \frac{m}{M l} u \\ \ddot{q}_2 &= -\frac{k_1}{M l} \dot{q}_1 - \frac{(M+m) k_2}{M m l^2} \dot{q}_2 + \frac{(M+m) g}{M l^2} q_2 + \frac{(M+m)}{M m l^2} u \end{cases}$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m g}{M} & -\frac{k_1}{M} & -\frac{k_2}{M l} \\ 0 & \frac{(M+m) g}{M l^2} & -\frac{k_1}{M l} & -\frac{(M+m) k_2}{M m l^2} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{m}{M l} \\ \frac{(M+m)}{M m l^2} \end{bmatrix}$$